

Let the dot product do the seeing

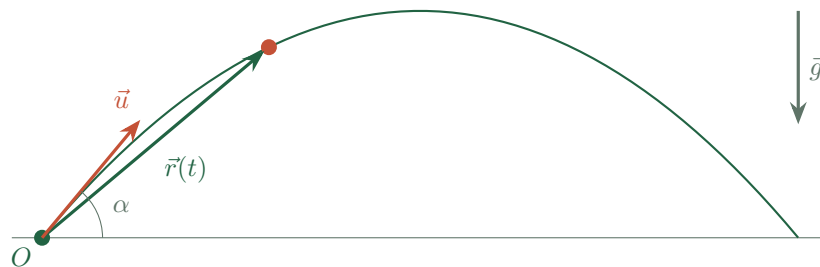
Exercise session 3 · Motion without coordinates

Solve using vector equations and dot products, without choosing coordinate axes or resolving vectors into components. Problems 1 and 2 are required; Problem 3 is an optional modeling challenge.

PROBLEM 1

A familiar flight, a different route

A projectile is launched with speed u at an angle α above the horizontal, where $0 < \alpha < 90^\circ$. Take the launch point O as the origin, and let $\vec{r}(t)$ be the projectile's position vector. Gravity is uniform, with acceleration $\vec{g}$ of magnitude g . Ignore air resistance. The projectile lands at the same height from which it was launched.



Using only $\vec{r}(t) \cdot \vec{r}(t)$ and its derivative with respect to time, determine:

- the maximum height above the launch point;
- the total time of flight.

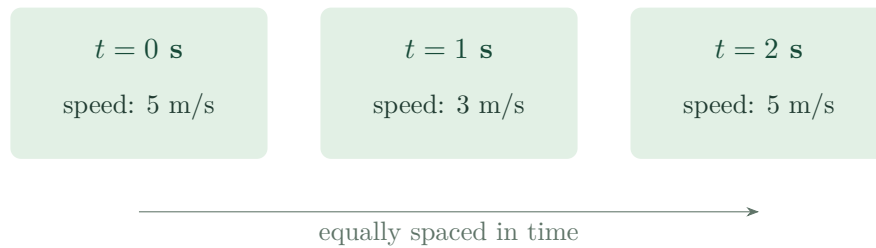
Optional reflection. At the highest point, is $\vec{r} \cdot \vec{r}$ stationary? Explain physically.

Working and explanation

PROBLEM 2

Three speeds, an unseen force

A 2 kg puck moves on a horizontal frictionless table under a constant horizontal net force $\vec{F}$. A sensor records only its **speed**, not its velocity direction or position. These are its readings:



- Find the magnitude of $\vec{F}$ and the angle it makes with the initial velocity. Can speed readings determine the force's absolute direction across the table?
- At what time between the first and last readings is the puck slowest? Find that speed. Does it ever stop during this interval?
- Find the straight-line distance between its positions at $t = 0$ and $t = 2$ s. What angle does this displacement make with $\vec{F}$?

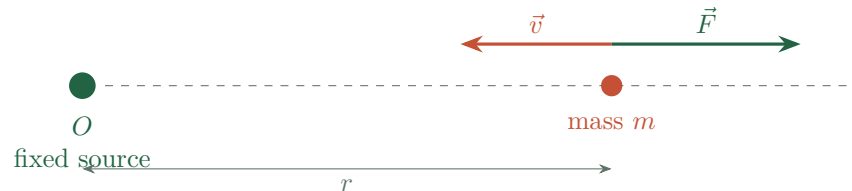
Working and explanation

PROBLEM 3

What force turned it back?

Optional · Modeling challenge

A particle of mass m approaches a fixed source O along a straight line. The only net force on the particle is a repulsion directed away from O ; its magnitude $F(r)$ depends only on the separation $r = |\vec{r}|$, where $\vec{r}$ is measured from O . The particle slows down, turns, and moves away without touching the source. Very far from O , its incoming speed approaches u .



The approach is described by the following idealized readings, where ℓ is a fixed length:

Separation r	4ℓ	2ℓ	ℓ
Squared speed v^2	$\frac{63}{64}u^2$	$\frac{7}{8}u^2$	0

- Starting from Newton's second law and a dot product with velocity, find a relation between $F(r)$ and the slope of v^2 as a function of r .
- Suppose the deficit $u^2 - v^2$ follows a power law in r . Use the readings to find a model for $v^2(r)$ and hence a closed expression for $F(r)$. Does doubling r halve the force?
- The same particle now approaches the same source with incoming speed $2u$ very far away. Predict its closest separation, assuming your force law remains valid. Is this prediction within the range of separations in the table, or does it extend the model beyond them?
- At the turning point, the instantaneous rate of change of v^2 is zero. Must the force also be zero? Explain.

Working and explanation
